

Project 1: Music Band Website

Requirements

[Coldplay official website](#) used as a reference. This was to ground the project within the context of real life. Using information from its website, I was able to populate the merchandise, songs, albums, and band_members tables. A [data generator](#) was used to populate customer tables, as well as any unknown data, as it would be impossible for me to obtain the real data of customers, as well as band_members address, due to privacy protection.

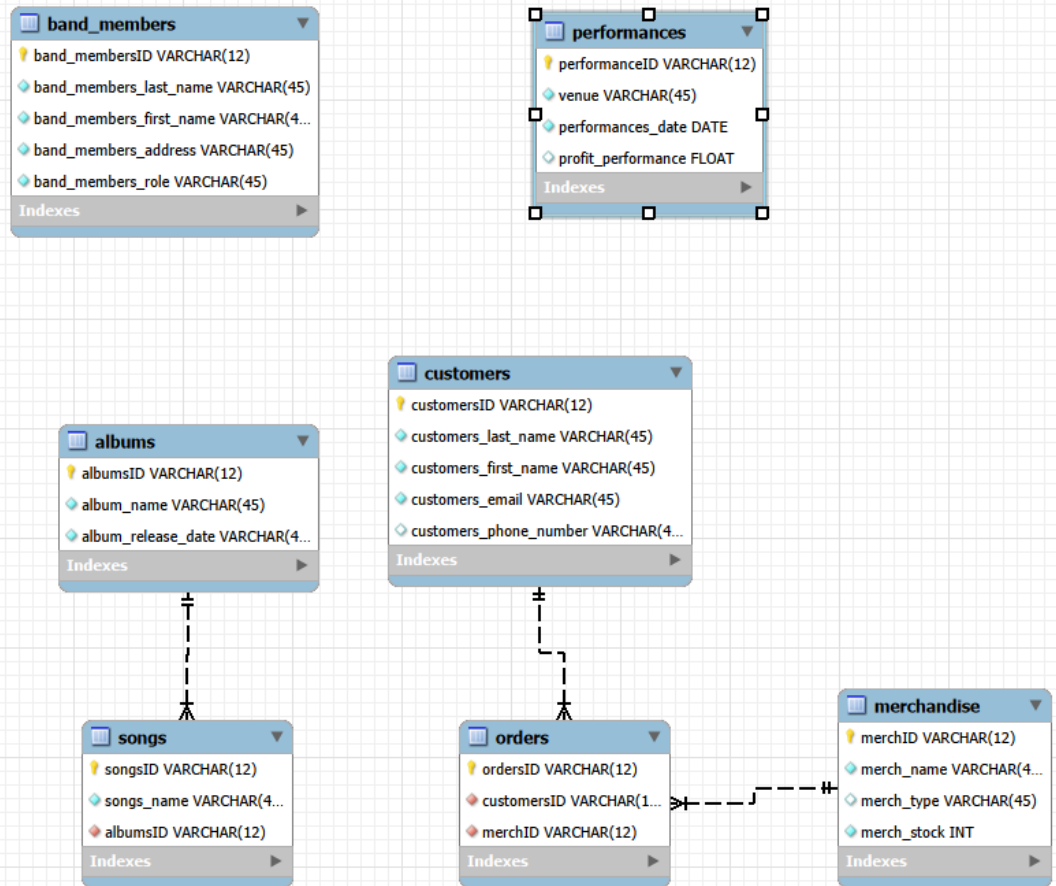
Entities(attributes)

band_members (band_membersID, band_members_last_name, band_members_first_name, band_members_address, band_members_role)
merchandise (merchID, merch_name, merch_type, merch_stock)
customers (customersID, customers_last_name, customers_first_name, customers_email, customers_phone_number, customers_address)
performances (performanceID, venue, performances_date, profit_performance)
albums (albumsID, album_name, album_release_date)
orders (ordersID, merchID, customersID)
songs (songsID, songs_name, albumsID)

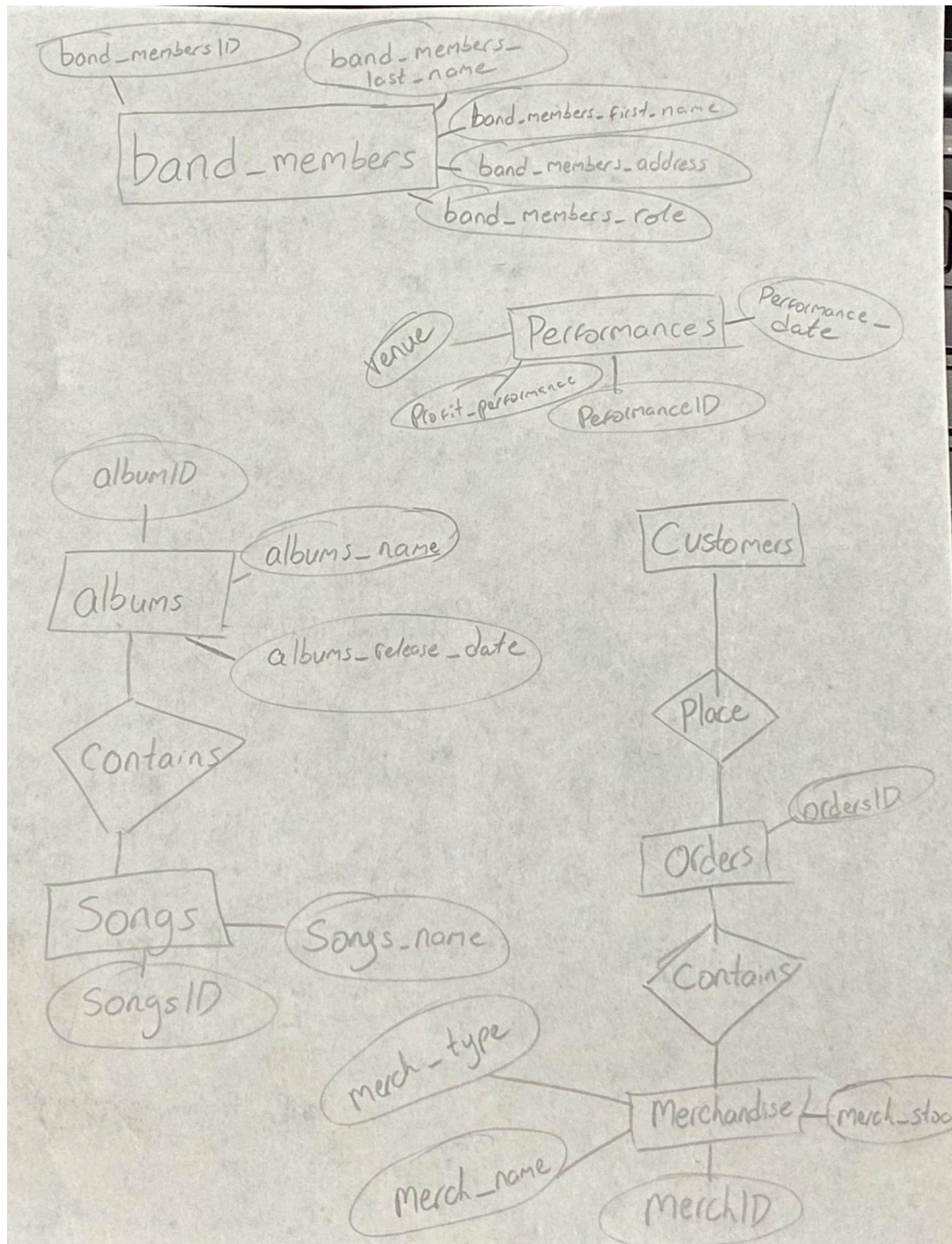
Steps

First, I identified entities and attributes, as seen above. Then, I drew an ER diagram using the Chen Notation. Afterwards, I used CREATE TABLE to create tables for each entity in MYSQL, as seen below in the SQL script. Afterwards, I used INSERT INTO to add data into the tables. At certain points, I had to use ALTER TABLE to change column names and constraints, for convenience and to remedy errors, respectively.

ER Diagram MYSQL



ER Diagram Chen Notation



SQL script (QUERIES)

-- TABLES CREATION

```
CREATE TABLE `band_website`.`band_members` (  
  `idband_members` INT NOT NULL,  
  `band_members_last_name` VARCHAR(45) NOT NULL,  
  `band_members_first_name` VARCHAR(45) NOT NULL,  
  `band_members_address` VARCHAR(45) NOT NULL,  
  `band_members_role` VARCHAR(45) NOT NULL,  
  PRIMARY KEY (`idband_members`),  
  UNIQUE INDEX `idband_members_UNIQUE` (`idband_members` ASC) VISIBLE);  
ALTER TABLE `band_website`.`band_members`  
CHANGE COLUMN `idband_members` `idband_members` VARCHAR(12) NOT NULL ;  
ALTER TABLE `band_website`.`band_members`  
CHANGE COLUMN `idband_members` `band_membersID` VARCHAR(12) NOT NULL ;
```

```
CREATE TABLE `band_website`.`merchandise` (  
  `merchID` INT NOT NULL,  
  `merch_name` VARCHAR(45) NOT NULL,  
  `merch_type` VARCHAR(45) NULL,  
  `merch_stock` INT NOT NULL,  
  PRIMARY KEY (`merchID`),  
  UNIQUE INDEX `merchID_UNIQUE` (`merchID` ASC) VISIBLE);  
ALTER TABLE `band_website`.`merchandise`  
CHANGE COLUMN `merchID` `merchID` VARCHAR(12) NOT NULL ;
```

```
CREATE TABLE `band_website`.`customers` (  
  `customersID` VARCHAR(12) NOT NULL,  
  `customers_last_name` VARCHAR(45) NOT NULL,  
  `customers_first_name` VARCHAR(45) NOT NULL,  
  `customers_email` VARCHAR(45) NOT NULL,  
  `customers_phone_number` VARCHAR(45) NULL,  
  PRIMARY KEY (`customersID`),  
  UNIQUE INDEX `customerID_UNIQUE` (`customersID` ASC) VISIBLE,  
  UNIQUE INDEX `customers_email_UNIQUE` (`customers_email` ASC) VISIBLE,  
  UNIQUE INDEX `customers_phone_number_UNIQUE` (`customers_phone_number` ASC) VISIBLE);  
ALTER TABLE `band_website`.`customers`  
ADD COLUMN `customers_address` VARCHAR(45) NOT NULL AFTER `customers_phone_number`;
```

```
CREATE TABLE `band_website`.`performances` (  
  `performanceID` VARCHAR(12) NOT NULL,  
  `venue` VARCHAR(45) NOT NULL,  
  `performances_date` DATE NOT NULL,  
  `profit_performance` FLOAT NULL,  
  PRIMARY KEY (`performanceID`),  
  UNIQUE INDEX `performanceID_UNIQUE` (`performanceID` ASC) VISIBLE);
```

```
CREATE TABLE `band_website`.`orders` (  
  `ordersID` VARCHAR(12) NOT NULL,  
  `customersID` VARCHAR(12) NOT NULL,  
  `merchID` VARCHAR(12) NOT NULL,  
  PRIMARY KEY (`ordersID`),  
  UNIQUE INDEX `ordersID_UNIQUE` (`ordersID` ASC) VISIBLE,  
  INDEX `customersID_idx` (`customersID` ASC) VISIBLE,  
  UNIQUE INDEX `customersID_UNIQUE` (`customersID` ASC) VISIBLE,
```

```

UNIQUE INDEX `merchID_UNIQUE` (`merchID` ASC) VISIBLE,
CONSTRAINT `customersID`
  FOREIGN KEY (`customersID`)
    REFERENCES `band_website`.`customers` (`customersID`)
    ON DELETE NO ACTION
    ON UPDATE NO ACTION,
CONSTRAINT `merchID`
  FOREIGN KEY (`merchID`)
    REFERENCES `band_website`.`merchandise` (`merchID`)
    ON DELETE NO ACTION
    ON UPDATE NO ACTION);

```

```

CREATE TABLE `band_website`.`albums` (
  `albumsID` VARCHAR(12) NOT NULL,
  `album_name` VARCHAR(45) NOT NULL,
  `album_release_date` VARCHAR(45) NOT NULL,
  PRIMARY KEY (`albumsID`),
  UNIQUE INDEX `albumsID_UNIQUE` (`albumsID` ASC) VISIBLE,
  UNIQUE INDEX `album_name_UNIQUE` (`album_name` ASC) VISIBLE);

```

```

CREATE TABLE `band_website`.`songs` (
  `songsID` VARCHAR(12) NOT NULL,
  `songs_name` VARCHAR(45) NOT NULL,
  `albumsID` VARCHAR(12) NOT NULL,
  PRIMARY KEY (`songsID`),
  UNIQUE INDEX `songsID_UNIQUE` (`songsID` ASC) VISIBLE,
  UNIQUE INDEX `songs_name_UNIQUE` (`songs_name` ASC) VISIBLE,
  UNIQUE INDEX `albumID_UNIQUE` (`albumsID` ASC) VISIBLE,
  CONSTRAINT `albumsID`
    FOREIGN KEY (`albumsID`)
      REFERENCES `band_website`.`albums` (`albumsID`)
      ON DELETE NO ACTION
      ON UPDATE NO ACTION);

```

-- REMOVING UNIQUE VALUE CONSTRAINT FROM FOREIGN KEY

```

ALTER TABLE songs
DROP FOREIGN KEY albumsID;
ALTER TABLE songs
DROP INDEX albumID_UNIQUE;
ALTER TABLE songs
ADD CONSTRAINT fk_songs_albums
FOREIGN KEY (albumsID)
REFERENCES albums(albumsID)
ON DELETE NO ACTION
ON UPDATE NO ACTION;

```

-- INSERTS

```

INSERT INTO band_members
VALUES ('0001', 'Martin', 'Chris', '15303 Ventura Blvd. Suite 900', 'Piano'),
      ('0002', 'Berryman', 'Guy', '88-90 Baker Street, London, W1U 6TQ', 'Bass'),
      ('0003', 'Champion', 'Will', '610 S Queen St Martinsburg WV 25401', 'Drums'),
      ('0004', 'Buckland', 'Jonny', 'Londons Belsize Park area', 'Guitar');

```

```

INSERT INTO customers
VALUES ('102351', 'Valdez', 'Zane', 'augue.porttitor@google.ca', '(867) 857-0541', '296-9269 Odio Ave'),
      ('731707', 'Donaldson', 'Joseph', 'tempus.lorem.fringilla@protonmail.ca', '1-382-741-2227', '436-8950 Mi Road'),
      ('538901', 'Contreras', 'Keaton', 'dolor.nonummy@aol.net', '(885) 944-6269', 'Ap #628-5820 Enim. St'),
      ('292872', 'Elliott', 'Kenneth', 'orci.ut.semper@protonmail.ca', '(889) 622-1943', 'Ap #997-729 Mauris Av'),

```

```
('260726','Barlow','Autumn','dignissim.pharetra@icloud.ca','1-906-416-1323','P.O. Box 978, 2965 Mauris Av');
```

INSERT INTO albums

```
VALUES ('1001','Parachutes','2000-07-10'),
('1002','Viva la Vida or Death and All His Friends','2008-06-12'),
('1003','Love in Tokyo','2018-12-07');
```

INSERT INTO merchandise

```
VALUES ('12001','EVERYONE IS AN ALIEN SOMEWHERE TEE','Shirt',568),
('12002','MUSIC OF THE SPHERES PLANET HOODIE','Sweater',233),
('12003','MOON MUSIC BRACELETS','Bracelet',780),
('12004','MOON MUSIC HAT','Hat',200),
('12005','VOLTIK TEE','Shirt',427);
```

INSERT INTO orders

```
VALUES ('34215','102351','12001'),
('03945','731707','12002'),
('73950','538901','12003'),
('48678','292872','12004'),
('99003','260726','12005');
```

INSERT INTO performances

```
VALUES ('6543','Wembley Stadium, United Kingdom','2025-09-12',124900),
('3456','Rogers Stadium, Canada','2025-07-11',80200),
('8008','Sun Bowl Stadium, USA','2025-06-14',205000),
('3932','Goyang Stadium, South Korea','2025-04-16',160000);
```

INSERT INTO songs

```
VALUES ('75614','Sparks','1001'),
('51039','The Scientist','1003'),
('86836','Yellow','1001'),
('23949','Parachutes','1001'),
('42181','Viva la Vida','1002'),
('82212','Paradise','1003'),
('27311','Yes','1002'),
('39543','Death and All His Friends','1002');
```

SQL (FORWARD ENGINEER)

```
-- MySQL Workbench Forward Engineering
```

```
SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0;
SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0;
SET @OLD_SQL_MODE=@@SQL_MODE,
SQL_MODE='ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION';
```

```
-- -----
-- Schema mydb
-- -----
```

```
-- Schema band_website
-- -----
```

```
-- -----
-- Schema band_website
-- -----
```

```
CREATE SCHEMA IF NOT EXISTS `band_website` DEFAULT CHARACTER SET utf8mb4 COLLATE utf8mb4_0900_ai_ci ;
USE `band_website` ;
```

```
-- Table `band_website`.`albums`
```

```
CREATE TABLE IF NOT EXISTS `band_website`.`albums` (  
  `albumsID` VARCHAR(12) NOT NULL,  
  `album_name` VARCHAR(45) NOT NULL,  
  `album_release_date` DATE NOT NULL,  
  PRIMARY KEY (`albumsID`),  
  UNIQUE INDEX `albumsID_UNIQUE` (`albumsID` ASC) VISIBLE,  
  UNIQUE INDEX `album_name_UNIQUE` (`album_name` ASC) VISIBLE)  
ENGINE = InnoDB  
DEFAULT CHARACTER SET = utf8mb4  
COLLATE = utf8mb4_0900_ai_ci;
```

```
-- Table `band_website`.`band_members`
```

```
CREATE TABLE IF NOT EXISTS `band_website`.`band_members` (  
  `band_membersID` VARCHAR(12) NOT NULL,  
  `band_members_last_name` VARCHAR(45) NOT NULL,  
  `band_members_first_name` VARCHAR(45) NOT NULL,  
  `band_members_address` VARCHAR(45) NOT NULL,  
  `band_members_role` VARCHAR(45) NOT NULL,  
  PRIMARY KEY (`band_membersID`),  
  UNIQUE INDEX `idband_members_UNIQUE` (`band_membersID` ASC) VISIBLE)  
ENGINE = InnoDB  
DEFAULT CHARACTER SET = utf8mb4  
COLLATE = utf8mb4_0900_ai_ci;
```

```
-- Table `band_website`.`customers`
```

```
CREATE TABLE IF NOT EXISTS `band_website`.`customers` (  
  `customersID` VARCHAR(12) NOT NULL,  
  `customers_last_name` VARCHAR(45) NOT NULL,  
  `customers_first_name` VARCHAR(45) NOT NULL,  
  `customers_email` VARCHAR(45) NOT NULL,  
  `customers_phone_number` VARCHAR(45) NULL DEFAULT NULL,  
  `customers_address` VARCHAR(45) NOT NULL,  
  PRIMARY KEY (`customersID`),  
  UNIQUE INDEX `customerID_UNIQUE` (`customersID` ASC) VISIBLE,  
  UNIQUE INDEX `customers_email_UNIQUE` (`customers_email` ASC) VISIBLE,  
  UNIQUE INDEX `customers_phone_number_UNIQUE` (`customers_phone_number` ASC) VISIBLE)  
ENGINE = InnoDB  
DEFAULT CHARACTER SET = utf8mb4  
COLLATE = utf8mb4_0900_ai_ci;
```

```
-- Table `band_website`.`merchandise`
```

```
CREATE TABLE IF NOT EXISTS `band_website`.`merchandise` (  
  `merchID` VARCHAR(12) NOT NULL,  
  `merch_name` VARCHAR(45) NOT NULL,  
  `merch_type` VARCHAR(45) NULL DEFAULT NULL,  
  `merch_stock` INT NOT NULL,  
  PRIMARY KEY (`merchID`),  
  UNIQUE INDEX `merchID_UNIQUE` (`merchID` ASC) VISIBLE)  
ENGINE = InnoDB
```

```
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;
```

```
-- Table `band_website`.`orders`
```

```
CREATE TABLE IF NOT EXISTS `band_website`.`orders` (
  `ordersID` VARCHAR(12) NOT NULL,
  `customersID` VARCHAR(12) NOT NULL,
  `merchID` VARCHAR(12) NOT NULL,
  PRIMARY KEY (`ordersID`),
  UNIQUE INDEX `ordersID_UNIQUE` (`ordersID` ASC) VISIBLE,
  INDEX `customersID_idx` (`customersID` ASC) VISIBLE,
  CONSTRAINT `customersID`
    FOREIGN KEY (`customersID`)
      REFERENCES `band_website`.`customers` (`customersID`),
  CONSTRAINT `merchID`
    FOREIGN KEY (`merchID`)
      REFERENCES `band_website`.`merchandise` (`merchID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;
```

```
-- Table `band_website`.`performances`
```

```
CREATE TABLE IF NOT EXISTS `band_website`.`performances` (
  `performanceID` VARCHAR(12) NOT NULL,
  `venue` VARCHAR(45) NOT NULL,
  `performances_date` DATE NOT NULL,
  `profit_performance` FLOAT NULL DEFAULT NULL,
  PRIMARY KEY (`performanceID`),
  UNIQUE INDEX `performanceID_UNIQUE` (`performanceID` ASC) VISIBLE)
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;
```

```
-- Table `band_website`.`songs`
```

```
CREATE TABLE IF NOT EXISTS `band_website`.`songs` (
  `songsID` VARCHAR(12) NOT NULL,
  `songs_name` VARCHAR(45) NOT NULL,
  `albumsID` VARCHAR(12) NOT NULL,
  PRIMARY KEY (`songsID`),
  UNIQUE INDEX `songsID_UNIQUE` (`songsID` ASC) VISIBLE,
  UNIQUE INDEX `songs_name_UNIQUE` (`songs_name` ASC) VISIBLE,
  INDEX `fk_songs_albums` (`albumsID` ASC) VISIBLE,
  CONSTRAINT `fk_songs_albums`
    FOREIGN KEY (`albumsID`)
      REFERENCES `band_website`.`albums` (`albumsID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;
```

```
SET SQL_MODE=@OLD_SQL_MODE;
SET FOREIGN_KEY_CHECKS=@OLD_FOREIGN_KEY_CHECKS;
```



```
SET UNIQUE_CHECKS=@OLD_UNIQUE_CHECKS;
```

```
INSERT INTO band_members
```

```
VALUES ('0001', 'Martin', 'Chris', '15303 Ventura Blvd. Suite 900', 'Piano'),
        ('0002', 'Berryman', 'Guy', '88-90 Baker Street, London, W1U 6TQ', 'Bass'),
        ('0003', 'Champion', 'Will', '610 S Queen St Martinsburg WV 25401', 'Drums'),
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        ('538901', 'Contreras', 'Keaton', 'dolor.nonummy@aol.net', '(885) 944-6269', 'Ap #628-5820 Enim. St'),
        ('292872', 'Elliott', 'Kenneth', 'orci.ut.semper@protonmail.ca', '(889) 622-1943', 'Ap #997-729 Mauris Av'),
        ('260726', 'Barlow', 'Autumn', 'dignissim.pharetra@icloud.ca', '1-906-416-1323', 'P.O. Box 978, 2965 Mauris Av');
```

```
INSERT INTO albums
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```
VALUES ('1001', 'Parachutes', '2000-07-10'),
        ('1002', 'Viva la Vida or Death and All His Friends', '2008-06-12'),
        ('1003', 'Love in Tokyo', '2018-12-07');
```

```
INSERT INTO merchandise
```

```
VALUES ('12001', 'EVERYONE IS AN ALIEN SOMEWHERE TEE', 'Shirt', 568),
        ('12002', 'MUSIC OF THE SPHERES PLANET HOODIE', 'Sweater', 233),
        ('12003', 'MOON MUSIC BRACELETS', 'Bracelet', 780),
        ('12004', 'MOON MUSIC HAT', 'Hat', 200),
        ('12005', 'VOLTIK TEE', 'Shirt', 427);
```

```
INSERT INTO orders
```

```
VALUES ('34215', '102351', '12001'),
        ('03945', '731707', '12002'),
        ('73950', '538901', '12003'),
        ('48678', '292872', '12004'),
        ('99003', '260726', '12005');
```

```
INSERT INTO performances
```

```
VALUES ('6543', 'Wembley Stadium, United Kingdom', '2025-09-12', 124900),
        ('3456', 'Rogers Stadium, Canada', '2025-07-11', 80200),
        ('8008', 'Sun Bowl Stadium, USA', '2025-06-14', 205000),
        ('3932', 'Goyang Stadium, South Korea', '2025-04-16', 160000);
```

```
INSERT INTO songs
```

```
VALUES ('75614', 'Sparks', '1001'),
        ('51039', 'The Scientist', '1003'),
        ('86836', 'Yellow', '1001'),
        ('23949', 'Parachutes', '1001'),
        ('42181', 'Viva la Vida', '1002'),
        ('82212', 'Paradise', '1003'),
        ('27311', 'Yes', '1002'),
        ('39543', 'Death and All His Friends', '1002');
```